

PHENYLSULFONAT CA

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Substance key: SXR004988

Revision Date: 09.12.2023

Version : 2 - 2 / T

Date of printing : 05.04.2026

1. PRODUCT AND COMPANY IDENTIFICATION

Trade name

PHENYLSULFONAT CA

Material number: 105575

Substance name: Calcium alkyl benzene sulphonate in isobutanol

Use of the Substance/Mixture

Industry sector : Crop Protection
Type of use: Agricultural Industry

Supplier name/address/telephone no.:

Clariant Produkte (Deutschland) GmbH

Frankfurt am Main 65926, Germany

Telephone : +49 69 305 18000

Information about the substance/preparation

BU Care Chemicals
Product Stewardship
E-mail address: SDS.Europe@clariant.com

Emergency telephone number: +65 3158 1074

2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids : Category 3

Acute toxicity (Dermal) : Category 4

Skin corrosion/irritation : Category 2

Serious eye damage/eye irritation : Category 1

Specific target organ toxicity - single exposure : Category 3 (Respiratory system, Central nervous system)

Short-term (acute) aquatic hazard : Category 3

Long-term (chronic) aquatic hazard : Category 2

GHS label elements

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Hazard pictograms

:



Signal word

: Danger

Hazard statements

: H226 Flammable liquid and vapour.
H312 Harmful in contact with skin.
H315 Causes skin irritation.
H318 Causes serious eye damage.
H335 May cause respiratory irritation.
H336 May cause drowsiness or dizziness.
H402 Harmful to aquatic life.
H411 Toxic to aquatic life with long lasting effects.

Precautionary statements

: P101 If medical advice is needed, have product container or label at hand.
P102 Keep out of reach of children.
P103 Read label before use.

Prevention:

P210 Keep away from heat/ sparks/ open flames/ hot surfaces.
No smoking.
P233 Keep container tightly closed.
P240 Ground/bond container and receiving equipment.
P241 Use explosion-proof electrical/ ventilating/ lighting equipment.
P242 Use only non-sparking tools.
P243 Take precautionary measures against static discharge.
P261 Avoid breathing mist or vapours.
P264 Wash skin thoroughly after handling.
P271 Use only outdoors or in a well-ventilated area.
P273 Avoid release to the environment.
P280 Wear protective gloves/ eye protection/ face protection.

Response:

P303 + P361 + P353 IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower.
P304 + P340 + P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/ physician.
P312 Call a POISON CENTER or doctor/ physician if you feel unwell.
P332 + P313 If skin irritation occurs: Get medical advice/ attention.
P362 Take off contaminated clothing and wash before reuse.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction.

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P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

P403 + P235 Store in a well-ventilated place. Keep cool.

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts	68953-96-8	>= 70 -< 90
Iso-Butanol	78-83-1	>= 30 -< 50

4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.
Do not leave the victim unattended.
- If inhaled : Consult a physician after significant exposure.
If unconscious, place in recovery position and seek medical advice.
- In case of skin contact : If skin irritation persists, call a physician.
If on skin, rinse well with water.
If on clothes, remove clothes.
- In case of eye contact : Small amounts splashed into eyes can cause irreversible tissue damage and blindness.
In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Continue rinsing eyes during transport to hospital.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do NOT induce vomiting.

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Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.

Most important symptoms and effects, both acute and delayed : irritant effects
corrosive effects
Causes skin irritation.
Causes serious eye damage.

Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : Alcohol-resistant foam
Carbon dioxide (CO₂)
Dry powder

Specific hazards during firefighting : Do not allow run-off from fire fighting to enter drains or water courses.
Sulphur dioxide

Hazardous combustion products : None

Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Use a water spray to cool fully closed containers.
For safety reasons in case of fire, cans should be stored separately in closed containments.

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation.
Wear suitable protective clothing.
Keep away sources of ignition.

Environmental precautions : Do not empty into drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up : Pick up with absorbent material (eg sand, sawdust). Rinse away rest with water
Can be landfilled or incinerated, when in compliance with local regulations.

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7. HANDLING AND STORAGE

Advice on protection against fire and explosion : Keep away from sources of ignition - No smoking.
Vapours may form explosive mixture with air.
Vapours may spread long distances and ignite.
Electrical equipment should be protected to the appropriate standard.

Advice on safe handling : Avoid formation of aerosol.

Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Take precautionary measures against static discharges.
Open drum carefully as content may be under pressure.
To avoid spills during handling keep bottle on a metal tray.
Dispose of rinse water in accordance with local and national regulations.

Further information on storage conditions : Keep container tightly closed and in a well-ventilated place.
After sampling, container should be sealed

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Iso-Butanol	78-83-1	TWA	50 ppm	ACGIH

Personal protective equipment

Respiratory protection : Use only in well-ventilated areas.
In the case of vapour formation use a respirator with an approved filter.
Equipment should conform to EN 14387
If the occupational exposure limits cannot be met, in exceptional cases suitable respiratory equipment should be worn only for a short period of time.
ABEK-P3-filter

Hand protection

Break through time : 480 min
Glove thickness : 0.7 mm
Remarks : Long-term exposure Impervious butyl rubber gloves

Break through time : 30 min
Glove thickness : 0.4 mm
For short-term exposure (splash protection): Nitrile rubber

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gloves.

These types of protective gloves are offered by various manufacturers. Please note the manufacturers' detailed statements, especially about the minimum thickness and the minimum breakthrough time. Consider also the particular working conditions under which the gloves are being used.

- Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.
- Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.
- Protective measures : Avoid contact with skin and eyes.
Do not inhale vapours
- Hygiene measures : Wash hands before breaks and at the end of workday.
When using, do not eat, drink or smoke.

9. PHYSICAL AND CHEMICAL PROPERTIES

- Appearance : liquid
- Colour : brown
- Odour : characteristic
- Odour Threshold : not determined
- pH : 7
Concentration: 3 %
- Melting point : no data available
- Boiling point : not determined
- Flash point : 57 °C
(999.5 hPa)
Method: DIN EN ISO 3679 (closed cup)
- Evaporation rate : not determined
- Self-ignition : no data available
- Upper explosion limit / upper : no data available

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flammability limit

Lower explosion limit / Lower
flammability limit : no data available

Vapour pressure : no data available

Relative vapour density : no data available

Density : ca. 1 g/cm³ (25 °C)

Solubility(ies)

Water solubility : insoluble

Partition coefficient: n-
octanol/water : Not applicable

Auto-ignition temperature : no data available

Decomposition temperature : The product does not contain any chemical groups which suggest self-reactive properties, nor is the estimated SADT less than 75 °C, nor is the exothermic decomposition energy higher than 300 J/g.

Viscosity

Viscosity, dynamic : ca. 16,000 mPa.s (25 °C)

Viscosity, kinematic : no data available

Explosive properties : Not explosive
Method: Expert judgementOxidizing properties : The substance or mixture is not classified as oxidizing.
Method: Expert judgementMetal corrosion rate : < 6.25 mm/a
Not corrosive to metals

Particle size : Not applicable

10. STABILITY AND REACTIVITY

Reactivity : No decomposition if stored and applied as directed.

Chemical stability : Stable under normal conditions.
No decomposition if stored and applied as directed.Possibility of hazardous
reactions : No dangerous reaction known under conditions of normal use.
Vapours may form explosive mixture with air.

Conditions to avoid : Heat, flames and sparks.

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Incompatible materials	:	Strong oxidizing agents
Hazardous decomposition products	:	When handled and stored appropriately, no dangerous decomposition products are known

11. TOXICOLOGICAL INFORMATION**Acute toxicity****Product:**

Acute oral toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401

Acute inhalation toxicity : Remarks: no data available

Acute dermal toxicity : LD50 (Rat, male and female): > 1,000 - 1,600 mg/kg
Method: OECD Test Guideline 402

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Acute oral toxicity : LD50 (Rat, male and female): > 2,000 mg/kg
Method: OECD Test Guideline 401
GLP: yes
Assessment: The substance or mixture has no acute oral toxicity

Acute inhalation toxicity : Remarks: not required

Acute dermal toxicity : LD50 (Rat, male and female): 1,000 - 1,600 mg/kg
Method: OECD Test Guideline 402
GLP: yes
Assessment: The component/mixture is moderately toxic after single contact with skin.

Iso-Butanol:

Acute oral toxicity : LD50 (Rat, male and female): 2,830 - 3,350 mg/kg
Method: OECD Test Guideline 401
GLP: yes

Acute inhalation toxicity : LC50 (Rat, male and female): > 18.18 mg/l
Exposure time: 6 h
Test atmosphere: vapour
Method: Other
GLP: yes
Assessment: The substance or mixture has no acute inhalation toxicity

Acute dermal toxicity : LD50 (Rabbit, male and female): 2,460 mg/kg
Method: OECD Test Guideline 402
GLP: yes

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Skin corrosion/irritation

Product:

Remarks : no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Species : Rabbit
Exposure time : 4 h
Assessment : Irritating to skin.
Method : Other
Result : Irritating to skin.
GLP : yes

Iso-Butanol:

Species : Rabbit
Exposure time : 4 h
Method : OECD Test Guideline 404
Result : Irritating to skin.
GLP : yes

Serious eye damage/eye irritation

Product:

Remarks : no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Species : Rabbit
Result : Risk of serious damage to eyes.
Exposure time : 1 h - 14 d
Assessment : Risk of serious damage to eyes.
Method : Other
GLP : yes

Iso-Butanol:

Species : Rabbit
Result : Risk of serious damage to eyes.
Exposure time : 24 h
Assessment : Risk of serious damage to eyes.
Method : OECD Test Guideline 405
GLP : yes

Respiratory or skin sensitisation

Product:

Test Type : Skin
Species : Guinea pig

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Method : OECD Test Guideline 406
Result : No sensitisation effects are known

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Test Type : Maximisation Test
Exposure routes : Dermal
Species : Guinea pig
Method : OECD Test Guideline 406
Result : Not a skin sensitizer.
GLP : yes

Assessment : Harmful in contact with skin., Causes skin irritation.,
Causes serious eye damage.

Iso-Butanol:

Exposure routes : Skin contact
Method : QSAR
Result : Not a skin sensitizer.

Assessment : Causes skin irritation., Causes serious eye damage.

Germ cell mutagenicity**Product:**

Germ cell mutagenicity - : Not mutagenic in Ames Test
Assessment

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 4 - 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster ovary cells
Concentration: 0,6 - 60 µg/ml
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes
Remarks: By analogy with a product of similar composition

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI

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Application Route: oral (gavage)
Exposure time: 72 h
Dose: 1122 mg/kg
Method: OECD Test Guideline 474
Result: negative
GLP: yes
Remarks: By analogy with a product of similar composition

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Iso-Butanol:

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Concentration: 100- 10000 µg/plate
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: No information available.

Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster lung cells
Concentration: 11 - 53 mM
Metabolic activation: without
Method: Other
Result: negative
GLP: No information available.

Test Type: In vitro gene mutation study in mammalian cells
Test system: Chinese hamster lung cells
Concentration: <= 107 mM
Metabolic activation: with and without metabolic activation
Method: Other
Result: negative
GLP: No information available.

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse (male and female)
Strain: NMRI
Cell type: Bone marrow
Application Route: oral (gavage)
Exposure time: single iapplication
Dose: 500 - 1000 - 2000 mg/kg
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Germ cell mutagenicity - Assessment : In vitro tests did not show mutagenic effects, In vivo tests did not show mutagenic effects

Carcinogenicity**Product:**

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Carcinogenicity -
Assessment : No information available.

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Carcinogenicity -
Assessment : No information available.

Iso-Butanol:

Carcinogenicity -
Assessment : No information available.

Reproductive toxicity**Product:**

Reproductive toxicity -
Assessment : No information available.

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Effects on fertility : Test Type: Three-generation study
Species: Rat, male and female
Strain: Other
Application Route: oral (feed)
Dose: 14 - 70 - 350 mg/kg bw d
Duration of Single Treatment: 720 d
General Toxicity - Parent: NOAEL: 350 mg/kg body weight
General Toxicity F1: NOAEL: 350 mg/kg body weight
General Toxicity F2: NOAEL: 350 mg/kg body weight
Method: Other
GLP: no
Remarks: By analogy with a product of similar composition

Effects on foetal
development : Test Type: Pre-natal
Species: Mouse, female
Strain: CD1
Application Route: oral (gavage)
Dose: 0,2 - 2,0 - 300 - 600 mg/kg
Duration of Single Treatment: 9 d
General Toxicity Maternal: NOAEL: 2 mg/kg body weight
Teratogenicity: NOAEL: 300 mg/kg body weight
Method: Other
GLP: No information available.
Remarks: By analogy with a product of similar composition

Iso-Butanol:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Strain: Sprague-Dawley

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Application Route: Inhalation
Dose: 500 - 1000 - 2500 ppm
Duration of Single Treatment: 6 h
General Toxicity - Parent: NOAEL: $\geq$ 7.5 mg/l
General Toxicity F1: NOAEL: $\geq$ 7.5 mg/l
General Toxicity F2: NOAEL: $\geq$ 7.5 mg/l
Method: OPPTS 870.3800
GLP: yes

Effects on foetal
development

: Test Type: Pre-natal
Species: Rat
Strain: wistar
Application Route: Inhalation
Dose: 0,5 - 2,5 - 10 mg/l
Duration of Single Treatment: 6 h
General Toxicity Maternal: NOAEL: 10 mg/l
Teratogenicity: NOAEL: 10 mg/l
Method: OECD Test Guideline 414
GLP: yes

Reproductive toxicity -
Assessment

: No reproductive toxicity to be expected.
No teratogenic effects to be expected.

STOT - single exposure**Product:**

Remarks : no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

Iso-Butanol:

Assessment : May cause respiratory irritation.
May cause drowsiness or dizziness.

STOT - repeated exposure**Product:**

Remarks : no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

Iso-Butanol:

Assessment : The substance or mixture is not classified as specific target

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organ toxicant, repeated exposure.

Repeated dose toxicity**Product:**

Remarks : no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Species : Rat, male and female
NOAEL : 50 mg/kg bw/day
LOAEL : 250 mg/kg
Application Route : oral (feed)
Exposure time : 84 d
Number of exposures : daily in feed
Dose : 50 - 250 mg/kg bw d
Control Group : yes
Method : Other
GLP : no
Remarks : By analogy with a product of similar composition

Species : Rat, male and female
NOAEL : 85 mg/kg bw/day
LOAEL : 145 mg/kg
Application Route : Drinking water
Exposure time : 270 d
Number of exposures : daily in trinking water
Dose : 85 - 145 - 430 mg/kg bw d
Control Group : yes
Method : Other
GLP : no
Remarks : By analogy with a product of similar composition

Repeated dose toxicity - Assessment : Harmful in contact with skin., Causes skin irritation., Causes serious eye damage.

Iso-Butanol:

Species : Rat, male and female
NOAEL : > 1450 mg/kg bw/day
Application Route : Drinking water
Exposure time : 90 d
Number of exposures : continuous
Dose : 1000 - 4000 - 16000 ppm
Control Group : yes
Method : OECD Test Guideline 408
GLP : yes

Species : Rat, male and female
NOAEL : ca. 3 mg/l
Application Route : Inhalation
Exposure time : 102 d

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Number of exposures : 6 hours/day, 5 days/week
Dose : 250 - 1000 - 2500 ppm
Control Group : yes
Method : Other
GLP : yes

Application Route : Skin contact
Remarks : not tested.

Repeated dose toxicity - Assessment : Causes skin irritation., Causes serious eye damage.

Aspiration toxicity**Product:**

no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

No aspiration toxicity classification

Iso-Butanol:

No aspiration toxicity classification

Further information**Product:**

Remarks : Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.
Concentrations substantially above the TLV value may cause narcotic effects.
Solvents may degrease the skin.

12. ECOLOGICAL INFORMATION**Ecotoxicity****Product:**

Toxicity to fish : LC50 (Danio rerio (zebra fish)): > 10 - 100 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 56 - 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : Remarks: no data available

Toxicity to microorganisms : Remarks: no data available

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Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 31.6 mg/l

End point: mortality

Exposure time: 96 h

Test Type: static test

Analytical monitoring: no

Method: OECD Test Guideline 203

GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 62 mg/l

End point: mortality

Exposure time: 48 h

Test Type: static test

Analytical monitoring: yes

Method: OECD Test Guideline 202

GLP: yes

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 29 mg/l

End point: Growth rate

Exposure time: 96 h

Test Type: static test

Analytical monitoring: no

Method: Other

GLP: no

Remarks: By analogy with a product of similar composition

NOEC (Pseudokirchneriella subcapitata (green algae)): 0.5 mg/l

End point: Growth rate

Exposure time: 96 h

Test Type: static test

Analytical monitoring: no

Method: Other

GLP: no

Remarks: By analogy with a product of similar composition

Toxicity to fish (Chronic toxicity) : NOEC (Oncorhynchus mykiss (rainbow trout)): 0.23 mg/l

End point: Other

Exposure time: 72 d

Test Type: flow-through test

Analytical monitoring: yes

Method: Other

GLP: yes

Remarks: By analogy with a product of similar composition

Toxicity to microorganisms : EC50 (activated sludge): 550 - 760 mg/l

End point: Bacteria toxicity (respiration inhibition)

Exposure time: 3 h

Test Type: aquatic

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Analytical monitoring: yes
Method: OECD Test Guideline 209
GLP: no data available

Toxicity to soil dwelling
organisms

: Test Type: artificial soil
LC50 (Eisenia fetida (earthworms)): > 1,000 mg/kg
Exposure time: 14 d
End point: mortality
Method: OECD Test Guideline 207
GLP: yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
NOEC (Eisenia fetida (earthworms)): 250 mg/kg
Exposure time: 14 d
End point: Body weight
Method: OECD Test Guideline 207
GLP: yes
Remarks: By analogy with a product of similar composition

Test Type: artificial soil
EC10 (Other): 82 - 100 mg/kg
Exposure time: 21 d
End point: Reproduction
Method: ISO 11267
GLP: no data available
Remarks: By analogy with a product of similar composition

Plant toxicity

: EC50: 167 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Guide-line 208
GLP: yes
Remarks: By analogy with a product of similar composition

NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Sorghum bicolor (sorghum)
Analytical monitoring: no
Method: OECD Guide-line 208
GLP: yes
Remarks: By analogy with a product of similar composition

EC50: 316 mg/kg
Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus
Analytical monitoring: no
Method: OECD Guide-line 208
GLP: yes
Remarks: By analogy with a product of similar composition

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NOEC: 100 mg/kg
Exposure time: 21 d
End point: Growth
Species: Phaseolus aureus
Analytical monitoring: no
Method: OECD Guide-line 208
GLP: yes
Remarks: By analogy with a product of similar composition

Sediment toxicity : Remarks: not tested.

Ecotoxicology Assessment

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

Iso-Butanol:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 1,430 mg/l
End point: mortality
Exposure time: 96 h
Test Type: flow-through test
Analytical monitoring: yes
Method: Other
GLP: no

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia pulex (Water flea)): 1,100 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: no

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 1,799 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to fish (Chronic toxicity) : Remarks: not required

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 20 mg/l
End point: Reproduction rate
Exposure time: 21 d
Analytical monitoring: yes
Method: Other
GLP: no
Remarks: The details of the toxic effect relate to the nominal concentration.

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Toxicity to microorganisms : IC50 (Natural microorganism): 670 mg/l
End point: Growth rate
Exposure time: 16 h
Test Type: static test
Analytical monitoring: no
Method: Other
GLP: No information available.

Persistence and degradability**Product:**

Biodegradability : Biodegradation: 10 - 30 %
Method: OECD Test Guideline 302B

Chemical Oxygen Demand (COD) : 1,230 mg/g

Dissolved organic carbon (DOC) : 340 mg/g

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Biodegradability : aerobic
Inoculum: activated sludge
Concentration: 22 mg/l
Carbon dioxide (CO₂)
Result: Not readily biodegradable.
Biodegradation: 2.9 %
Exposure time: 28 d
Method: OECD Test Guideline 301E
GLP: no

Physico-chemical removability : Remarks: Not readily biodegradable.

Stability in water : Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C) pH: 4
Hydrolysis: (16.7 %)
Method: OECD Test Guideline 111
GLP: yes

Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50 °C) pH: 7
Hydrolysis: (25.6 %)
Method: OECD Test Guideline 111
GLP: yes

Test Type: abiotic
Degradation half life (Aqueous buffer solution.): > 365 d (50

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°C) pH: 9
Hydrolysis: (18.6 %)
Method: OECD Test Guideline 111
GLP: yes

Iso-Butanol:

Biodegradability : aerobic
Inoculum: activated sludge
BOD in % of theoretical OD
Result: Readily biodegradable.
Biodegradation: 70 - 80 %
Exposure time: 28 d
Method: OECD Test Guideline 301D
GLP: no

Bioaccumulative potential**Product:**

Bioaccumulation : Remarks: no data available

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Bioaccumulation : Species: Other
Bioconcentration factor (BCF): 3.16
Method: Other
GLP: no
Remarks: Does not bioaccumulate.
The value is given based on a SAR/AAR approach using
OECD Toolbox, DEREK, VEGA QSAR models (CAESAR
models), etc.

Partition coefficient: n-octanol/water : log Pow: 4.595 (20 °C)
Method: Regulation (EC) No. 440/2008, Annex, A.8
GLP: yes

Iso-Butanol:

Bioaccumulation : Remarks: Due to the low logPow bioaccumulation is not expected

Partition coefficient: n-octanol/water : log Pow: 1 (25 °C)
Method: OECD Test Guideline 117
GLP: yes

Mobility in soil**Product:**

Distribution among environmental compartments : Remarks: no data available

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Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Distribution among environmental compartments : adsorption
Medium: Soil
log Koc: > 5
Method: OECD Test Guideline 121
Remarks: Based upon the calculated log Koc, adsorption to the soil phase is not expected.

Iso-Butanol:

Distribution among environmental compartments : Adsorption/Soil
Medium: water - soil
log Koc: 0.47
Method: estimated

Other adverse effects**Product:**

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life with long lasting effects.

Components:**Benzenesulfonic acid, mono-C11-13-branched alkyl derivs., calcium salts:**

Environmental fate and pathways : not available

Results of PBT and vPvB assessment : The substance is not identified as a PBT or as a vPvB substance.

Iso-Butanol:

Environmental fate and pathways : not available

Results of PBT and vPvB assessment : This substance is not considered to be persistent, bioaccumulating and toxic (PBT).

Additional ecological information : Do not allow to enter ground water, waterways or waste water.

13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.

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Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.
Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION

IATA

Proper shipping name: Flammable liquid, n.o.s., solution
Class: 3
Packing group: III
UN/ID number: UN 1993
Primary risk: 3
Remarks: Shipment permitted
Hazard inducer(s): Isobutyl alcohol
Benzenesulfonic acid, mono-C11-13-branched alkyl derivs.,
calcium salts

IMDG

Proper shipping name: Flammable liquid, n.o.s., solution
Class: 3
Packing group: III
UN no. UN 1993
Primary risk: 3
Remarks: Shipment permitted
Hazard inducer(s): Isobutyl alcohol
Hazard inducer/Marine pollutant: Benzenesulfonic acid, mono-C11-13-branched alkyl derivs.,
calcium salts
Marine pollutant: Marine Pollutant
EmS : F-E S-E

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

European Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures.

Observe national specific workplace risk prevention and health regulations.

Notification of the ministry of industry on the system of classification and hazard communication of hazardous substances B.E. 2555 (GHS 2009)

16. OTHER INFORMATION

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Further information

Other information : Observe national and local legal requirements

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Full text of other abbreviations

ACGIH : USA. ACGIH Threshold Limit Values (TLV)

ACGIH / TWA : 8-hour, time-weighted average

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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applicable law. You should obtain and review the applicable Material Safety Data Sheet information before handling any of these products. For additional information, please contact Clariant.

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